tighter, but for these big holes 1 degree is the best the network can do. In another ten years, when an interferometric detector is operating on the Moon, the error boxes will be reduced in size along one side by a factor of 100.

Because the holes' orbit was elongated, the computer concludes that the two holes were captured into orbit around each other only a few hours before they coalesced and emitted the burst. (If they had been orbiting each other for longer than a few hours, the push of gravitational waves departing from the binary would have made their orbit circular.) Recent capture means the holes were probably in a dense cluster of black holes and massive stars at the center of some galaxy.

The computer therefore examines catalogs of optical galaxies, radio galaxies, and X-ray galaxies, searching for any that reside in the 1 -degree error box, are between 0.8 and 1.2 billion light-years from Earth, and have peculiar cores. Forty candidates are found and turned over to astronomers. For the next few years these forty candidates will be studied in detail, with radio, millimeter, infrared, optical, ultraviolet, X-ray, and gamma-ray telescopes. Gradually it will become clear that one of the candidate galaxies has a core in which a massive agglomerate of gas and stars was beginning, when the light we now see left the galaxy, a million-year-Iong phase of violent evolution—an evolution that will trigger the birth of a gigantic black hole, and then a quasar. Thanks to the burst of gravitational waves which identified this specific galaxy as interesting, astronomers can now begin to unravel the details of how gigantic black holes are born.

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1. The factor $1/50$ comes from detailed calculations with the Einstein field equation. It includes a factor $1/(2\pi)$, which is approximately $1/6$, to convert the hole's circumference into a radius, and an additional factor $1/5$ that arises from details of the Einstein field equation.

2. Their lasers actually produced microwaves (short-wavelength radio waves) rather than light, and thus were called *masers* rather than "lasers." "Real" lasers, the kind that produce light, were not successfully constructed until several years later.

3. Braginsky has a remarkable mastery of the nuances of the English language; he can construct an eloquent English phrase to describe a new idea far more readily than most Americans or Britons.

4. A key foundation for our idea came from a colleague, William Unruh, at the University of British Columbia. The development of the idea and its consequences was carried out jointly by Caves, me, and three others who were gathered around the lunch table with us when we discovered it: Ronald Drever, Vernon Sandberg, and Mark Zimmermann.

5. The full idea is described by Caves et al. (1980) and by Braginsky, Vorontsov, and Thorne (1980).